1 Modulbeschreibung Advanced AI-Assisted Software Engineering

1	**Modulname** Advanced AI-Assisted Software Engineering
1.1	**Modulkürzel** A3SE
1.2	**Art** Wahlpflicht
1.3	**Lehrveranstaltung** Advanced AI-Assisted Software Engineering
1.4	**Semester X**
1.5	**Modulverantwortliche(r)** Prof. Dr. Bernhard Humm
1.6	**Weitere Lehrende** Prof. Dr. Bernhard Humm, Prof. Dr. Markus Voß
1.7	**Studiengangsniveau** Master
1.8	**Lehrsprache** Englisch
2	**Inhalt** The module covers the following topics: 1. Introduction • AI as a tool and co-engineer in modern software engineering • Overview of AI-assisted development workflows and agentic coding 2. Requirements & Analysis • AI-supported elicitation of functional and non-functional requirements • Personas, user stories, use cases • Wireframing and UI concept generation with AI tools • Documentation practices 3. Architecture & Design • Business architecture: entities, processes, use-case structures • Technical architecture: system components, interfaces, integration layers • AI-assisted architectural decision-making 4. Coding & Implementation • AI-assisted development of a full-stack web application (DB, backend, frontend) • Building chatbots and workflow automations • Introduction to multi-agent systems and agent frameworks 5. Testing • AI-generated test cases and test automation • Quality assurance and validation of AI-generated code

6. Re-Engineering

- AI-assisted inventory analysis and reverse engineering
- Restructuring, impact analysis and implementation planning
- AI-assisted forward-engineering
- Validation

7. Infrastructure & Operations

- Infrastructure as Code
- CI/CD concepts and pipelines
- Deployment considerations

8. Security

- Vulnerabilities introduced by AI-generated code
- AI-assissted threat modeling
- Regulatory and compliance security
- Security of autonomous agents

8. AI & Society

- Ethical, legal, and societal implications of AI in software engineering

Responsible use of AI tools in professional practice

3 Ziele

Upon successful completion of this module, students will be able to:

Knowledge & Understanding

- Explain the principles, opportunities, and limitations of AI-assisted software engineering and agentic development workflows.
- Understand how modern AI tools support requirements engineering, architecture design, coding, testing, and operations.
- Describe key concepts of business architecture, technical architecture, data modelling, and workflow automation.

Practical Skills

- Elicit, structure, and document software requirements using AI-supported techniques (personas, user stories, use cases, wireframes).
- Design business and technical architectures with AI assistance, including entity models and integration patterns.
- Implement a full-stack software system (database, backend, frontend) using AI-assisted coding tools.
- Build conversational agents, workflow automations, and simple multi-agent systems using modern AI frameworks.
- Generate and evaluate test cases with AI support.
- Re-engineer existing brownfield applications, including inventory analysis, reverse engineering, restructuring, forward engineering and validation.
- Apply basic DevOps practices such as Infrastructure as Code and CI/CD pipelines.

Cognitive & Methodological Competencies

- Awareness of opportunities, but also limitations of AI assisted software engineering, related to the current state-of-the art of tools as well as in principle.
- Critically assess AI-generated artefacts for correctness, completeness, and ethical implications.
- Collaborate effectively in teams using AI tools as co-engineers and reflect on new collaboration practices.
- Reflect on societal, ethical, and professional issues related to AI-assisted development.

4	**Lehr- und Lernformen** 6 SWS VP : integrierte Vorlesung (V) / Übung (Ü)
5	**Arbeitsaufwand und Credit Points** 9CP Präsenz (Vorlesung/Praktikum): 81 h (30 %) Vor-/Nachbereitung: 54 h (20 %) Projektarbeit: 81 h (30 %) Prüfungsvorbereitung: 54 h (20%) Gesamt: 270 h (100 %, 9CP)
6	**Prüfungsform, Prüfungsdauer und Prüfungsvoraussetzung** Mündliche Prüfung, keine Prüfungsvoraussetzung.
7	**Notwendige Kenntnisse** -
8	**Empfohlene Kenntnisse** -

9	**Dauer, zeitliche Gliederung und Häufigkeit des Angebots** 1 Semester, jährlich
10	**Verwendbarkeit des Moduls** • M.Sc. Informatik • M.Sc. Informatik Dual • M.Sc. Data Science,
11	**Literatur** • Ian Sommerville. Software Engineering. Tenth edition. Pearson, 2016. • Evans, E. *Domain-Driven Design.* Addison-Wesley. • Bass, L., Clements, P., Kazman, R. *Software Architecture in Practice.* Addison-Wesley. • Chris Birchall: Re-Engineering Legacy Software. Manning, 2016 • Mohamad Abou Ali et al. "Agentic AI: a comprehensive survey of architectures, applications, and future directions". In: Artificial Intelligence Review 59.1 • Alenezi, Mamdouh, and Mohammed Akour. 2025. "AI-Driven Innovations in Software Engineering: A Review of Current Practices and Future Directions" Applied Sciences 15, no. 3: 1344. • Bandi, Ajay, Bhavani Kongari, Roshini Naguru, Sahitya Pasnoor, and Sri Vidya Vilipala. 2025. "The Rise of Agentic AI: A Review of Definitions, Frameworks, Architectures, Applications, Evaluation Metrics, and Challenges" Future Internet 17, no. 9: 404. • Matteo Esposito, et al. "Generative AI for software architecture. Applications, challenges, and future directions". In: Journal of Systems and Software 231, S. 112607. Soodeh Hosseini und Hossein Seilani. "The role of agentic AI in shaping a smart future: A systematic review". In: Array 26, S. 100399.